

STAGE 7 VERIFICATION REPORT: SECTION-BY-SECTION GEMINI REPORT GENERATION

Verification Date: September 29, 2026 | Evaluation Suite: Automated PyTest v9.1.1, Next.js Build, Multi-Tenant IDOR & Gemini Live API | Overall Status: **VERIFIED & LOCKED (73/73 TESTS PASSED)**

EXECUTIVE VERDICT: STAGE 7 PRODUCTION READY & FULLY LOCKED (PASS)

Stage 7 of ARM (AI Academic Report Assistant) has been fully implemented, systematically tested, and locked. Stage 7 realizes the **Section-by-Section AI Report Content Generation Layer** powered by the Google Gemini model. Operating under strict non-negotiable constraints, Stage 7 generates high-fidelity report content section by section while enforcing: (1) Mandatory Stage 4 locked template schema; (2) Mandatory Stage 5 project facts and evidence metadata; (3) Mandatory Stage 6 approved report plan (rejecting unapproved or stale plans); (4) Zero-hallucination factual grounding with deterministic extraction and validation; (5) Robust prompt-injection defense via XML sandbox isolation; (6) Multi-tenant IDOR protection across jobs, drafts, and version histories; (7) Real-time generation job progress tracking; and (8) Versioned report persistence in public.reports and public.report_versions. In strict compliance with Stage 7 boundaries, this layer produces **content only** (no citation verification, external web research, image generation, DOCX/PDF rendering, or broad architecture rewrites). All 73 tests across Stages 1–7 pass with 0 regressions.

1. Comprehensive Test Suite Execution Summary

A full regression pass was executed across all testing suites. All 73 tests passed with zero failures and zero regressions against Stages 1–6.

Test Suite Module	Target Area / Layer	Tests	Result
test_stage7_report_generation.py	Gemini Section Generation, Validator, IDOR, Versioning, Injection Defense, Live API	9	PASS (9/9)
test_stage6_report_planner.py	AI Report Planner, Structural Alignment, Stale Detection, Plan Approval	7	PASS (7/7)
test_stage5_project_evidence.py	Project Info, Members, Evidence Locker, SHA-256 Checksums, Readiness	8	PASS (8/8)
test_stage4_template_review_lock.py	Template Review, Canonical Normalization, Lock Immutability	10	PASS (10/10)
test_stage3_template_intelligence.py	Deterministic AST Parsing, Formatting Extraction, Schema Gen	9	PASS (9/9)
test_stage2_template_ingestion.py	DOCX/PDF Upload, Validation, Storage Fingerprinting, Queue	10	PASS (10/10)
test_stage1_verification.py	Database Foundation, RLS, Schema, Multi-Tenant Isolation	7	PASS (7/7)
test_foundation_services.py	Storage Abstraction, AI Provider Mock/Real, Error Handling	4	PASS (4/4)
test_health.py	Health Checks, v1 System Diagnostics, Root Endpoints	3	PASS (3/3)
test_arm_database_foundation.py	Core Connection Pooling & PostgREST Integrity	1	PASS (1/1)
test_api_endpoints.py	Baseline API Routers, Project CRUD, Legacy Dispatch	3	PASS (3/3)
test_template_parser.py	Standalone Parser Benchmark & Fallback Schemas	1	PASS (1/1)
test_document_assembler.py	Document Assembler DOCX Rendering Pipeline	1	PASS (1/1)
TOTAL SUITE VERIFICATION	All System Subsystems & Stages 1–7	73	PASS (73/73)

2. Stage 7 Dedicated Test Suite Details (test_stage7_report_generation.py)

Test Case Name	Target Behavioral Invariant & Verification Scope	Result
test_locked_template_and_approved_plan_enforcement	Enforces that report generation cannot proceed without a locked template (HTTP 422) and an approved report plan (HTTP 422).	PASS
test_stale_plan_protection	Verifies that when project info is modified after plan approval, the plan is marked stale and generation is blocked until re-planning.	PASS
test_multitenant_idor_protection	Guarantees strict multi-tenant isolation; User B cannot trigger generation, read drafts, inspect jobs, or read versions of User A (HTTP 403/404).	PASS
test_deterministic_validator_unit	Direct unit validation verifying section ID match, block type normalization, meta-commentary removal, ungrounded citation stripping, and metric auditing.	PASS
test_section_by_section_generation_mock	Verifies section-by-section generation with isolated prompts, schema conforming outputs, progress tracking, and report persistence.	PASS
test_versioning_and_regeneration	Validates regeneration workflow: increments version_number from 1 to 2, preserves version 1 history in report_versions, and updates current pointer.	PASS
test_prompt_injection_defense	Verifies prompt injection attempts embedded in project descriptions or evidence files are neutralised via XML data sandboxing and factual validation.	PASS

test_different_project_inputs_and_anti_repetition	Tests distinct inputs (Attendance vs Agriculture); verifies domain-distinct generated text, absence of canned boilerplate repetition, and appropriate structure.	PASS
test_live_gemini_integration	Executes end-to-end section generation with live Gemini LLM; verifies structured JSON output adheres to ContentBlock schema and grounded facts.	PASS

3. Stage 7 Architectural Components & Deterministic Controls

Component / Module	Implementation Role & Strict Guardrails	Status
packages/report_generator/schema.py	Defines Pydantic schemas: BlockType, GroundingStatus, ContentBlock, MissingInformationItem, VisualRequirement, SectionGenerationOutput, GenerationMetadata, GeneratedReport.	VERIFIED
packages/report_generator/validator.py	SectionContentValidator: Deterministic post-generation validation. Checks section IDs, normalizes block types, strips meta-commentary ('Sure, here is...'), strips unverified evidence citations, detects metric hallucinations, preserves missing info & visual placeholders.	VERIFIED
packages/report_generator/generator.py	AIReportGenerator: Section-by-section generator. Enforces XML sandbox isolation (, ,), context windowing, structured output parsing with retries, and deterministic fallback synthesis.	VERIFIED
apps/api/services/report_generation.py	ReportGenerationService: Verifies project ownership, Stage 4 lock, Stage 6 approved plan; creates and updates generation_jobs with real section steps; manages versioning in reports and report_versions tables.	VERIFIED
apps/api/routers/projects.py	REST API endpoints: POST generate-report, GET report, GET report/versions, GET report/versions/{ver}, GET report/jobs/{id}. Fully enforces IDOR security.	VERIFIED
apps/web/src/components/arm/report-generator.tsx	Interactive React / Next.js client with generation button, live job progress indicator, section navigation, rich block renderer (headings, tables, callouts), grounding badges, and version switcher.	VERIFIED

4. Architectural Invariants & Stage 8 Readiness

Invariant	Verification Evidence & Downstream Contract
Content-Only Scope	Stage 7 strictly generates semantic report content. Citations are verified against registered files only; external web search, image generation, DOCX/PDF rendering, and final visual formatting are strictly deferred to subsequent stages.
Zero-Hallucination & Grounding	SectionContentValidator audits numerical claims and unverified evidence citations. Items lacking evidence are catalogued in missing_information_items rather than confabulated.
Deterministic Progress & Versioning	Generation progress updates generation_jobs row with real percent progress per section. Subsequent generation runs store snapshots in report_versions and advance version_number without destroying history.
Frontend Verification	apps/web passes TypeScript check (0 errors), ESLint (0 errors, 0 warnings), and Next.js static production build generates 26 static routes with full type-safety.
Stage 8 Readiness	The generated report data structure conforms exactly to GeneratedReport with validated ContentBlock primitives, ready for Stage 8 citation verification, research augmentation, or document rendering.